NOT Gate in VHDL

1. VHDL Code

```
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
  port (
    a : in  std_logic;
    y : out std_logic
  );
end not_gate;

architecture behavioral of not_gate is
begin
  y <= not a;
end behavioral;
```

2. Circuit

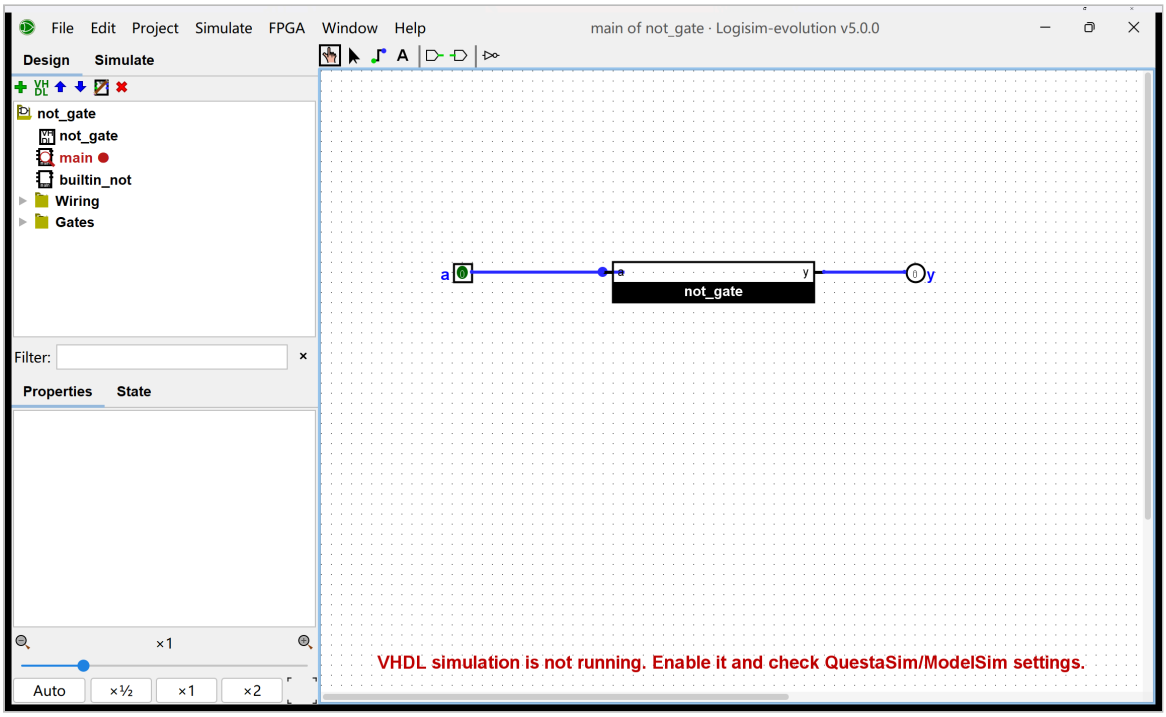


Figure 1. not_gate.vhd imported into Logisim-evolution (circuit main).

3. Simulation Results

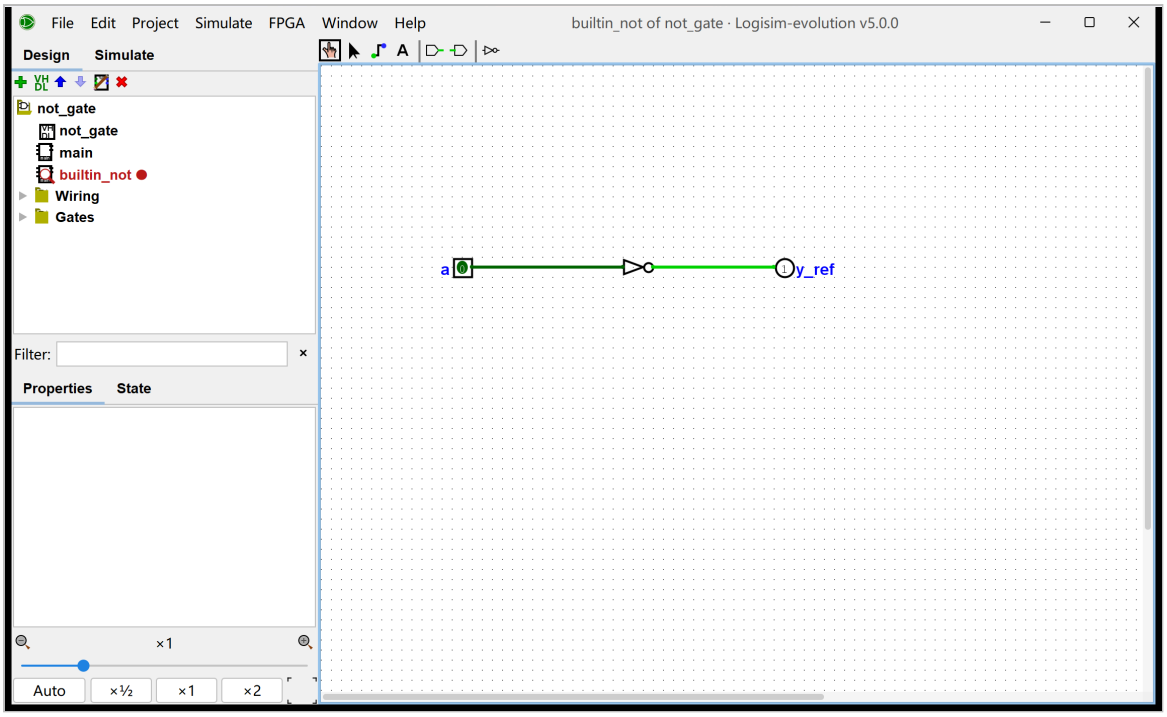


Figure 2. a = 0 → y = 1 (Logisim NOT gate).

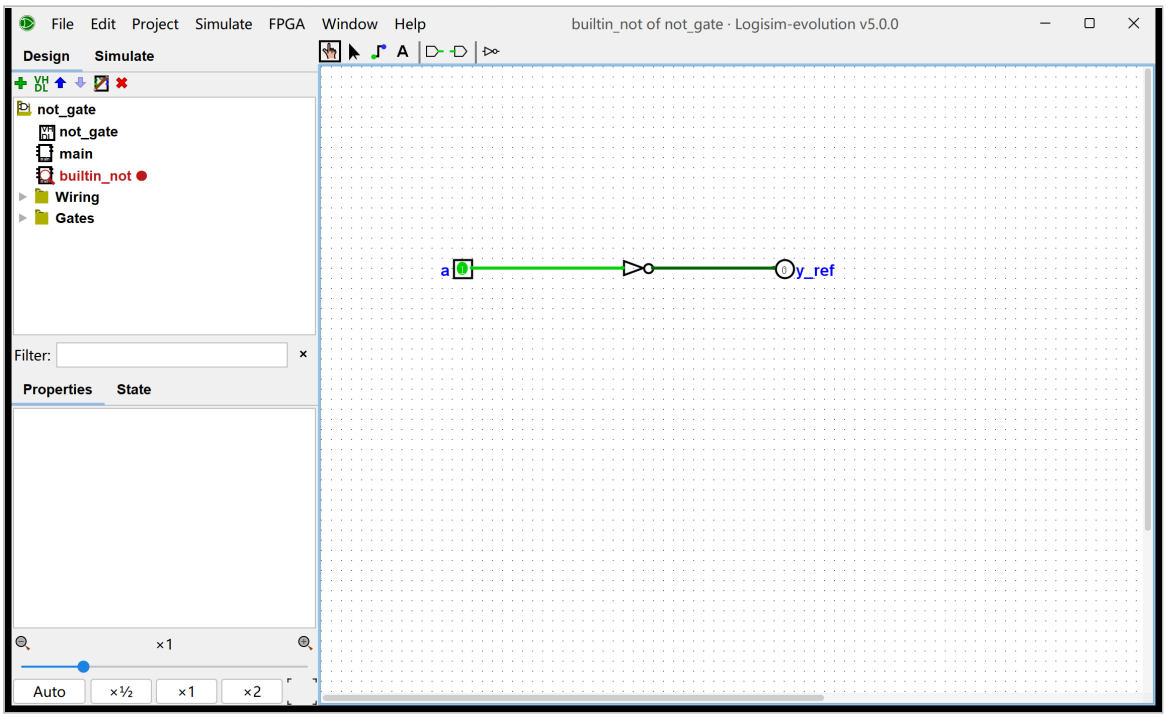


Figure 3. a = 1 → y = 0 (Logisim NOT gate).

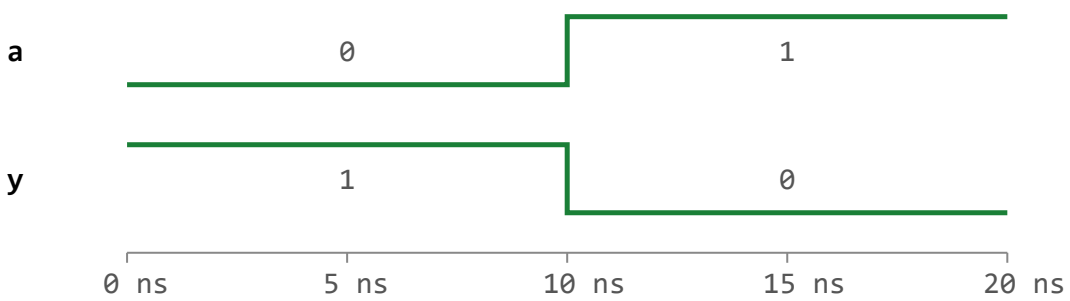


Figure 4. Simulation waveform of not_gate.vhd.

4. Explanation

a	y
0	1
1	0

The statement `y <= not a;` makes the output the inverse of the input: when a = 0 the output is y = 1, and when a = 1 the output is y = 0, matching the NOT gate truth table.